

WELDGUARD AI

Laser Welding Quality Inspection & Defect Analysis Certificate

REPORT ID: INSP-5238E53A

DATE: 2026-09-21 09:49 UTC

ISO 5817 Level B Advisory Protocol

Component ID:	TEST-CRACK-SPECIMEN-01	Inspection Date:	2026-09-21 09:49:58.423267
Operator:	Test Runner	Input Source:	DEMO
Material:	Inconel 718	Welding Process:	Fiber Laser Welding
Inspection Engine:	WeldGuard-CV-Precision-v2.4 v2.4.1	Data Classification:	DEMO SAMPLE DATA

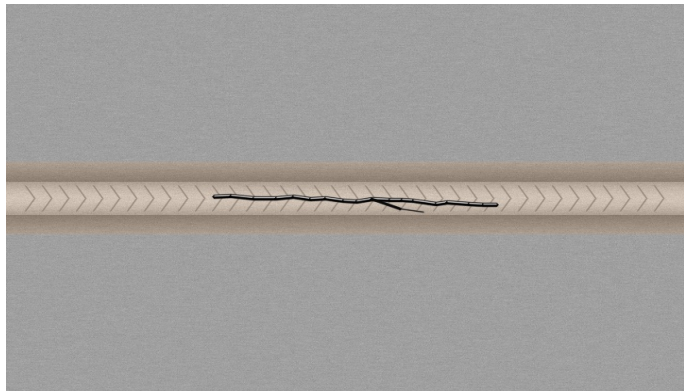
Defect Classification	Model Confidence	Defensible Severity	Human Review Status
Crack	96.0%	High	Approved

Defensible Severity Assessment Methodology

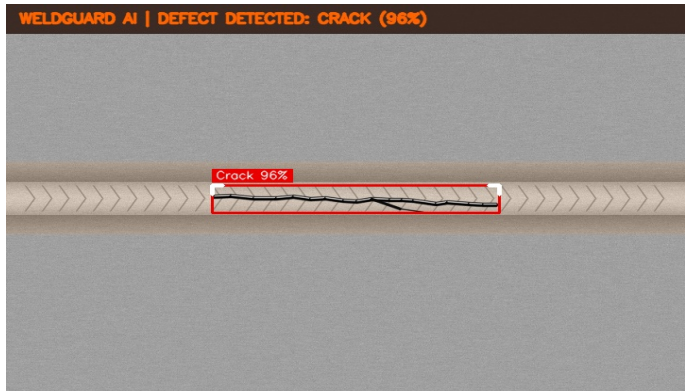
Basis of Severity Rating: Geometric Rubric (Criticality Tier 1): Cracks exhibit extreme stress concentration factors and propagation tendencies under dynamic cyclic load. Any confirmed crack is evaluated as High severity per ISO 5817 Level B standards.

Visual Inspection Evidence & Defect Localization

Original Raw Frame



Computer Vision Localized Defect Seam



Inspector Disposition & Review Notes

Current Review Status: Approved

Review Notes: Manually verified by senior NDT tech.

LEGAL & TECHNICAL DISCLAIMER: WeldGuard AI is an automated decision-support and computer vision advisory tool developed as a software prototype. Surface optical analysis does not inspect internal volumetric structures beneath the visible seam surface. All automated defect classifications and severity ratings must be corroborated by certified Non-Destructive Testing (NDT Level II/III) inspection personnel before releasing mission-critical welded assemblies. WeldGuard AI does not directly control physical welding machinery or modify live welding parameters.